Leonard Vincent Ramil

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LeoRamill

Education

2025–2026 **Double MSc in Computer Science**, *EURECOM*, Sophia Antipolis, France, *17.36/20*
Focus: Digital Image and Sound Music Processing, Optimization, Reinforcement Learning, Cloud Computing.
Semester Project: Quality Control of large database (Vessel Verse) annotations and model re-training.
Major: Data Science

2024–2026 **MSc in Data Science and Engineering**, *Politecnico di Torino*, Turin, Italy
Focus: Distributed System, Data Science, Information Theory, ML & Deep learning, Math for ML.

2021–2024 **BSc in Computer Science and AI**, *Sapienza University of Rome*, Italy, *110/110 cum laude*
Honours Program: Top 6 students. Focus on Safety, Robustness, Privacy & Explainability in AI, under supervision of Prof. Iacopo Masi.

2016–2021 **High School Diploma (Scientific Focus)**, *Liceo Statale "Camillo Cavour"*, Rome, Italy, *90/100*

Experience

March 2026 – Aug 2026 **Research Intern**, *Inria (EPIONE Team)*, Sophia Antipolis, France
Conducted research on scalable spatio-temporal analysis of multi-modal medical images (MRI/PET) under the supervision of Prof. Marco Lorenzi. The project focuses on extending Neural Cellular Automata architectures to predict neurodegenerative disease evolution, leveraging disentanglement learning to model the joint variability of structural and molecular brain data.

Sept 2024 – June 2025 **Data AI Engineer**, *CLUSTER Reply*, Turin, Italy
Engineered an Azure RAG vector database to optimize chatbot precision and developed GenAI models for automated document chunking, scaling the system to 10k+ daily documents with 25% higher efficiency.

May 2024 – Oct 2024 **Research Intern**, *HERCOLE Lab (Sapienza University of Rome)*, Rome, Italy
Conducted research on *Counterfactual Explainable AI for Anomaly Detection using Graph Neural Networks* applied to financial transactions, supervised by Prof. Gabriele Tolomei. Extended the CARLA library with GNN-based models (GNN, GAT, GIN), achieving a 20% improvement in validity and fidelity metrics within a PNRR project in collaboration with Intesa San Paolo.

Selected Projects

Sep 2025 – Dec 2025 **VesselVerse Quality Control**, *Python, PyTorch*
Built a multimodal network combining ResNet branches on three MIP projections with an MLP on 19 graph metrics to flag low-quality cerebrovascular segmentations, reaching ∼83% validation accuracy (vs. 79% tabular baseline), with GradCAM used to audit failure cases.

May 2025 – Jul 2025 **ADACS: Adversarial Domain Adaptation via Cross-Domain**, *Python, PyTorch, Wandb*
Developed an unsupervised domain adaptation framework aligning synthetic (GTA5) to real-world (Cityscapes) data. Improved mIoU from 26.38% to 37.01% (ResNet-101) and 23.80% to 29.76% (ResNet-18).

May 2024 – Oct 2024 **CARLA – Counterfactual And Recourse LibrAry**, *Python, PyTorch*
Extended CARLA benchmarking library by implementing novel graph-based counterfactual explainability methods as part of thesis work.

Jul 2023 – Sep 2023 **KYC - Know Your Cards**, *Python, Tensorflow*
Designed CV pipeline using bitmasking and CNNs to grade Pokémon TCG cards, achieving 92% classification accuracy on a test set of 500 images.

Publications

2026 Projection-Based Quality Assessment of Cerebrovascular Segmentations
Ion, B., Zhang, X., **Ramil, L. V.**, Boccadifuoco, A., Ourselin, S., Cleary, J., and Zuluaga, M. A.
SWITCH+, MICCAI Workshop on Stroke and Neurovascular Diseases, 2026

2026 PET-SMART: An Interpretable, Flexible, and Scalable Model of tau-PET Progression in Alzheimer's Disease
Ramil, L. V., Kalkhof, J., Humbert, O., and Lorenzi, M.
Under submission at MLCN, MICCAI Workshop on Machine Learning in Clinical Neuroimaging

Extracurricular Activity

Aug 2024 **BOOST'24 Summer School**, *ELICSIR Foundation*, Bologna, Italy
Intensive program on cutting-edge CS research with lectures from Cornell University and Sapienza faculty.

2022 – 2023 **Scholarship Team Lead**, *Sapienza Università di Roma*, Rome, Italy
Managed a tutor group assisting 200+ students, improving administrative response times by 15%.

Skills & Languages

Programming Python, Java, Go, C++, SQL, LaTeX, HTML, CSS, JavaScript, Vue.js

Tools Git, Docker, Figma, PyTorch, Tensorflow, Wandb

Languages **Italian** (Native), **English** (Professional), **Filipino** (Basic), **French** (A2)